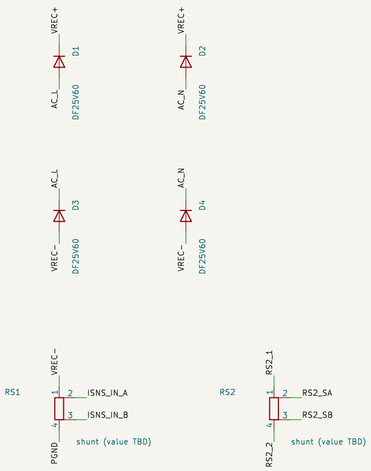
LeafOBC-WafflePlate: first-pass schematic of the 2012 Nissan Leaf OBC power module ("waffle plate")

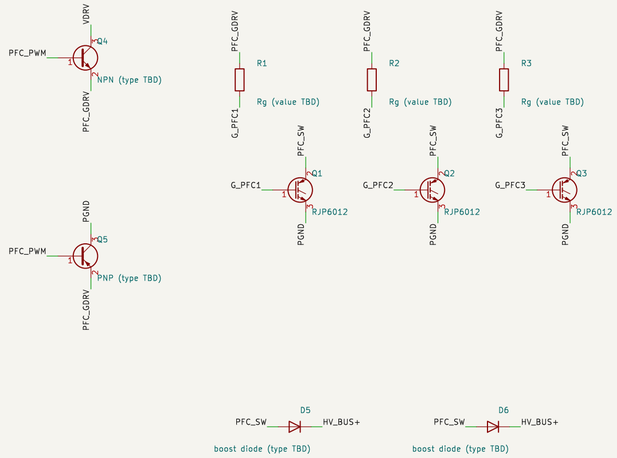
DRAFT. Topology is from the community draft schematic plus identified part numbers. Connections are NOT yet verified against the module. Hidden fields per part: Verified, Draft_ref, Location_px (pixel position in images/stitched/waffle-plate-top.jpg).

A. Input bridge and shunts (upper left)



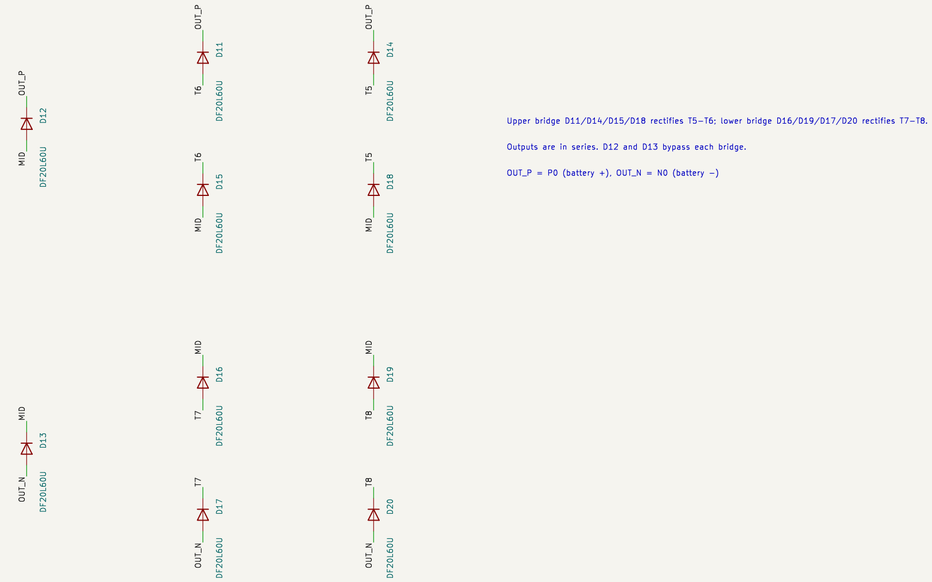
RS2: function and connections TBD

B. PFC boost stage (mid left)



Q1–Q3 in parallel. D5, D6 in parallel.

D. Output rectifier: two series-stacked bridges (upper right)

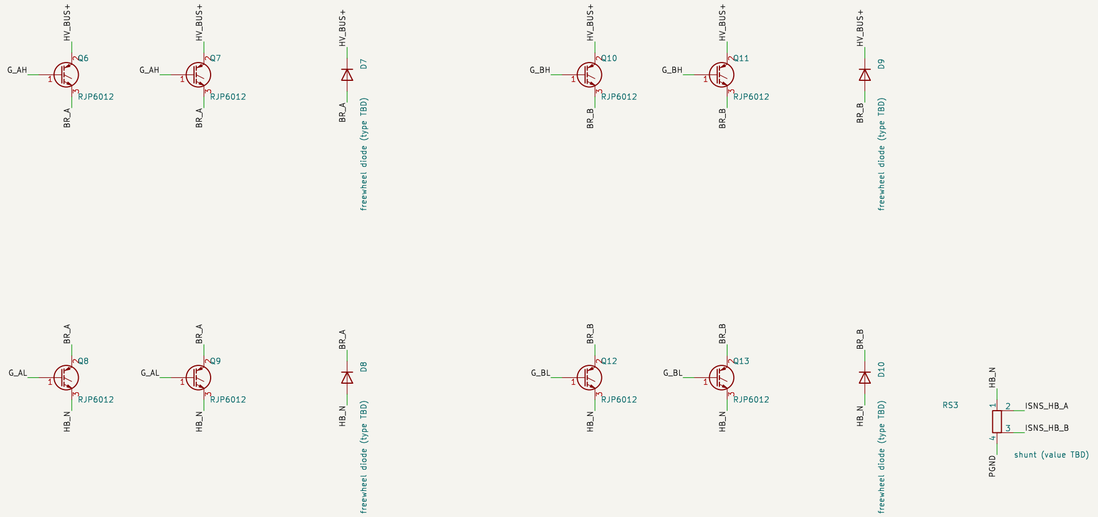


Upper bridge D11/D14/D15/D18 rectifies T5–T6; lower bridge D16/D19/D17/D20 rectifies T7–T8.

Outputs are in series. D12 and D13 bypass each bridge.

OUT_P = P0 (battery +). OUT_N = N0 (battery –).

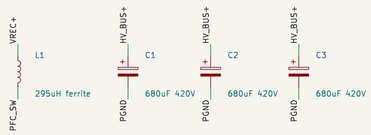
C. Primary full bridge, 2 IGBTs in parallel per switch (lower right)

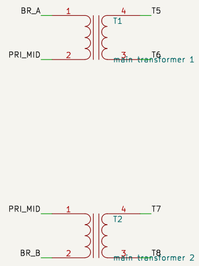


Gate and gate–emitter resistors around the bridge are not yet catalogued.

Which IGBT pair is which switch is a placeholder until traced.

E. Off-module parts (charger PCB)





Primaries shown in series: verify.

F. Module pads (draft numbering, unverified)

AC_L	P1	J1 L	HV_BUS+	P14	J27 P
AC_N	P2	J4 N	HV_BUS+	P15	J28 P
AC_N	P3	J5 N	ISNS_HB_A	P16	J32 (16/15)
VREC+	P4	J7 L1A	BR_A	P17	p23 T1
ISNS_IN_A	P5	J10 (5)	BR_B	P18	p25 T4
ISNS_IN_B	P6	J11 (8/7)	OUT_N	P19	p5 N0
PFC_PWM	P7	J14 (12/13)	T7	P20	p7 T7
PGND	P8	J18 N1	T8	P21	p10 T8
PGND	P9	J19 N1	T6	P22	p12 T6
PFC_SW	P10	J21	T5	P23	p14 T5
PFC_SW	P11	J22	OUT_P	P24	p16 P0
HV_BUS+	P12	J24 P1	FRAME_GND	P25	p1 FG5/FG6
HV_BUS+	P13	J25 P1			

Unassigned draft pads: J8 (6, IC312_PFC), J12/J13 (11), J15 (10/9), J30/J31 (N-N1), p27, p29, p31, p32, p33

CERN-OHL-S-2.0
First pass. Unverified connections.
LeafOBC-WafflePlate
Sheet: /
File: waffle-plate.kicad_sch

Title: Leaf OBC waffle plate: original module

Size: A2	Date:	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1